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Supporting Information for

NRLMSIS 2.1: An empirical model of nitric oxide incorporated into MSIS

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Introduction

The supporting information consists of additional tables and figures, Excel spreadsheets containing the model parameter values, and a Fortran software package for running the model.

Table S1. Residual Bias[a]

Data Set	150–200 km	120–150 km	105–120 km	100–105 km	95–100 km	90–95 km	85–90 km	80–85 km	75–80 km	70–75 km
Envisat/MIPAS	0.031	-0.006	-0.054	-0.032	0.129	0.277	0.239	0.170	0.225	0.271
ACE/FTS			-0.303	0.063	0.204	0.176	0.113	0.116	0.096	0.014
AIM/SOFIE		0.057	-0.168	-0.103	-0.096	-0.152	-0.218	-0.187	-0.063	-0.053
SNOE	-0.027	0.029	0.136	-0.038	-0.139					
Odin/SMR			0.250	0.143	-0.016	-0.005	-0.151	-0.428	-0.943	-1.148
UARS/HALOE		-0.058	-0.074	-0.005	-0.075	-0.119	-0.173	-0.061	0.033	0.019
Envisat/SCIAM.	0.742	0.405	-0.045	-0.132	-0.190	-0.277	-0.297	-0.091	0.032	0.138

[a]Shown are mean data-minus-model residuals in natural log space (see section 5 of the main text for details), for the indicated altitude bins and including all 15 analysis ensembles, as well as all of the Envisat/SCIAMACHY daily averages.

Table S2. Residual Standard Deviation[a]

Data Set	150–200 km	120–150 km	105–120 km	100–105 km	95–100 km	90–95 km	85–90 km	80–85 km	75–80 km	70–75 km
Envisat/MIPAS	0.653	0.740	0.767	0.817	0.813	0.838	0.894	0.980	1.050	1.057
ACE/FTS			0.755	0.682	0.706	0.752	0.787	0.806	0.751	0.671
AIM/SOFIE		0.972	0.865	0.884	0.927	1.002	1.088	1.102	1.099	1.033
SNOE	0.615	0.563	0.465	0.542	0.707					
Odin/SMR			0.563	0.558	0.581	0.593	0.668	0.639	0.622	0.815
UARS/HALOE		0.763	0.705	0.704	0.844	0.853	0.830	0.826	0.801	0.716
Envisat/SCIAM.	0.763	0.776	0.670	0.616	0.708	0.883	0.938	0.961	0.877	0.857

[a]Shown are the standard deviations of data-minus-model residuals in natural log space, for the indicated altitude bins and including all 15 analysis ensembles, as well as all of the Envisat/SCIAMACHY daily averages.

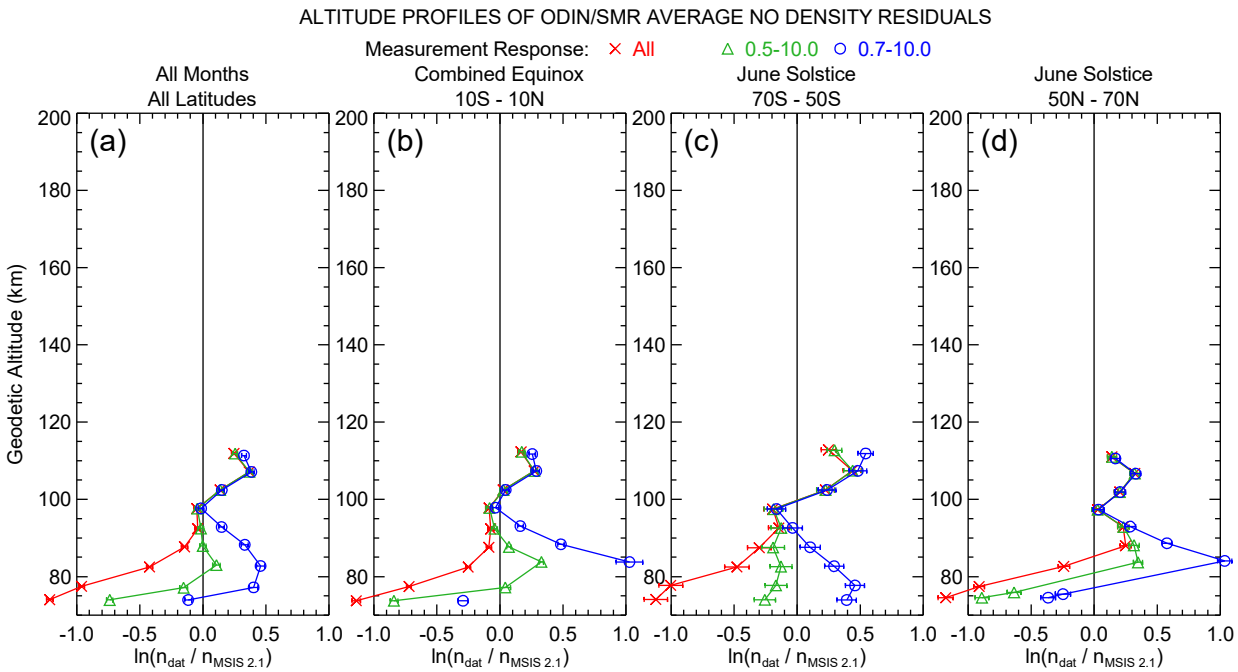

Figure S1. Average Odin/SMR-minus-MSIS 2.1 NO density residuals (in natural log space) as a function of altitude, for different selection criteria of the SMR measurement response: (red x's) all measurement response values included; (green triangles) measurement values < 0.5 excluded; (blue circles) values < 0.7 excluded. Retrievals with low measurement response values are more strongly influenced by the a priori NO density. The analysis is the same as shown for Odin/SMR in Figures 6a–d in the main text, except that the entire SMR data set was used here, instead of the analysis ensembles. Results are shown for the seasonal-latitudinal bins indicated above each panel.

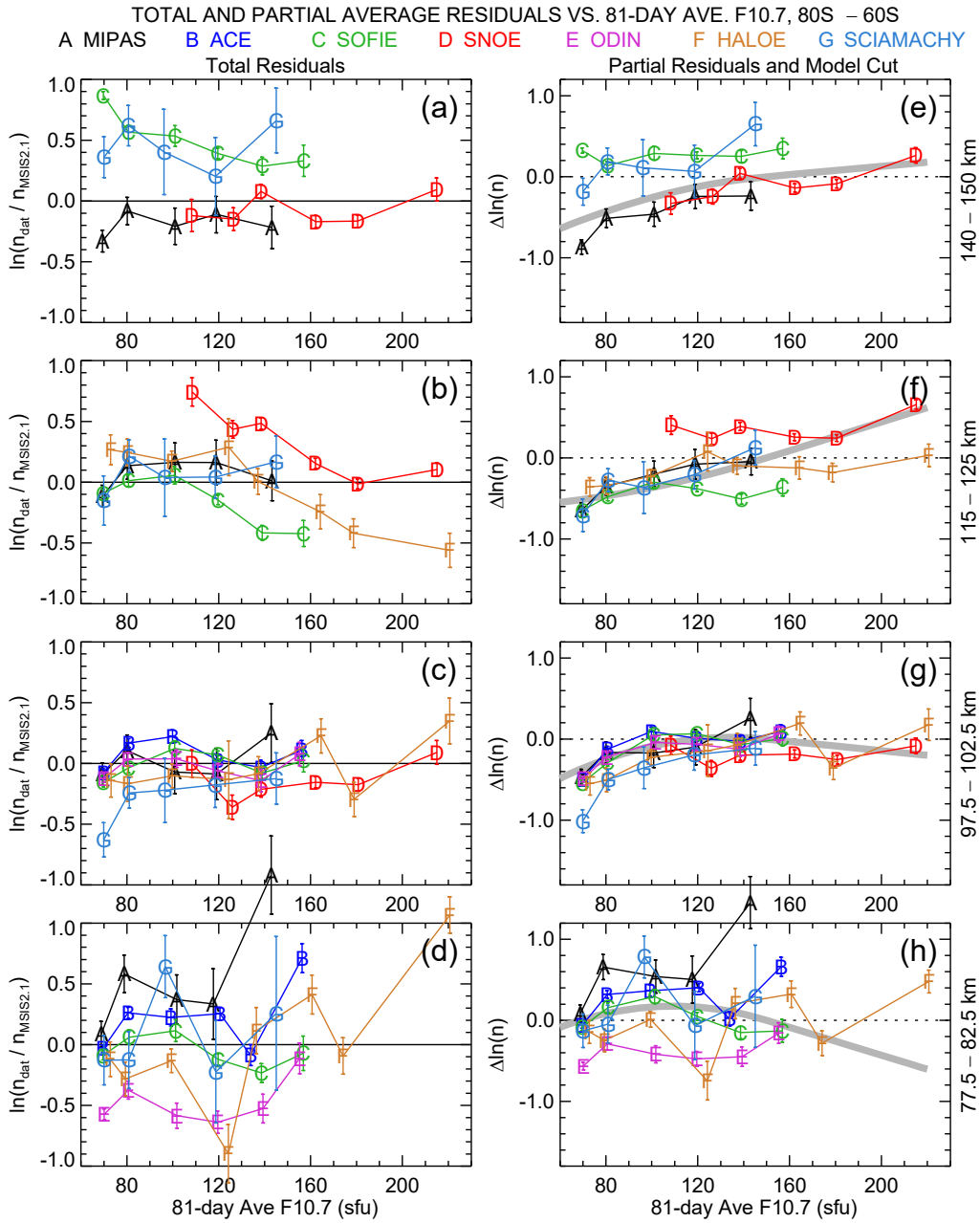

Figure S2. Total (left) and partial (right) data-minus-model residuals as a function of $\bar{F}_{10.7}$ (the 81-day average of the daily $F_{10.7}$). Same as Figure 9 in the main text, but showing results for high southern latitudes (80°S–60°S).

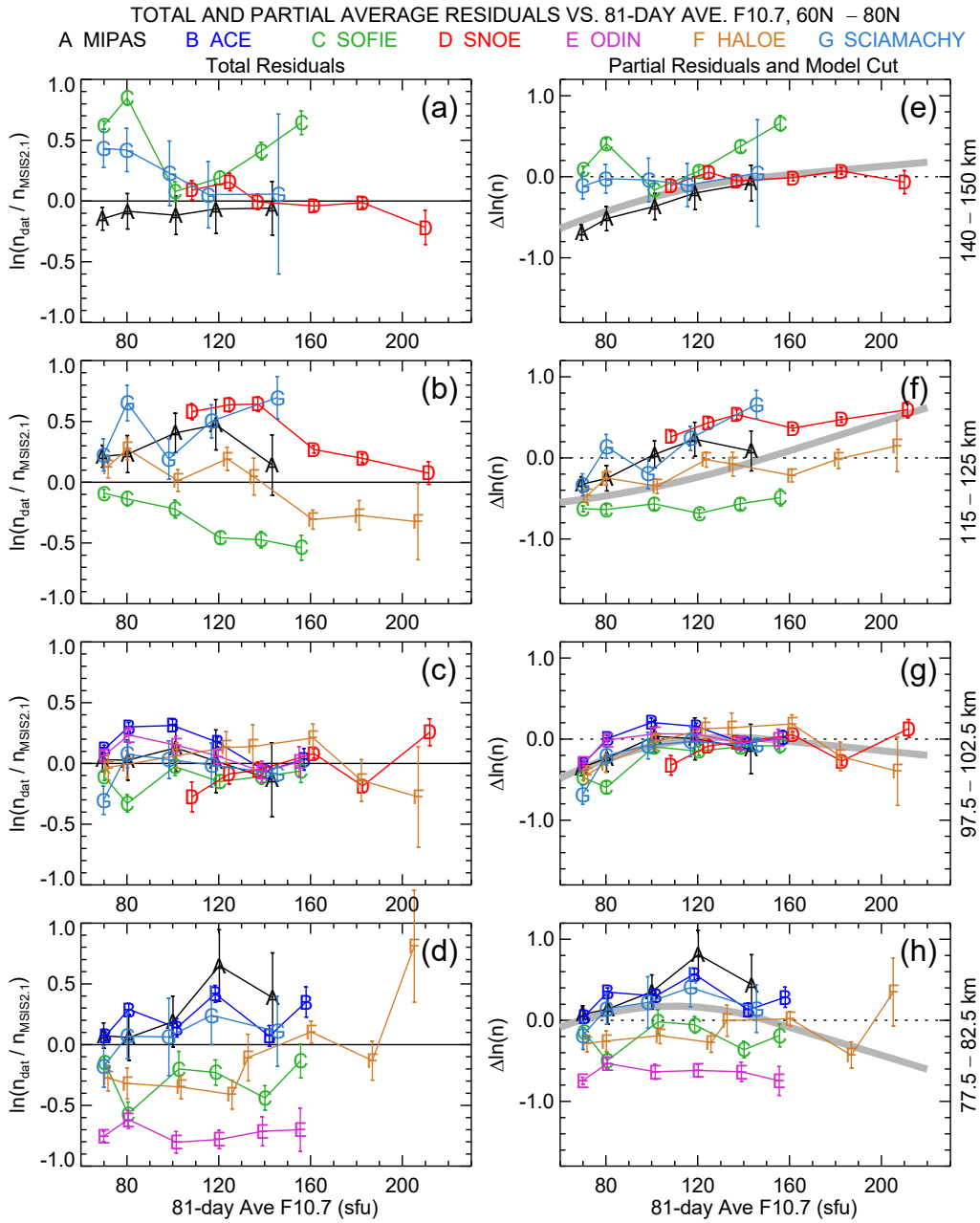

Figure S3. Total (left) and partial (right) data-minus-model residuals as a function of $\bar{F}_{10.7}$ (the 81-day average of the daily $F_{10.7}$). Same as Figure 9 in the main text, but showing results for high northern latitudes (60°N–80°N).

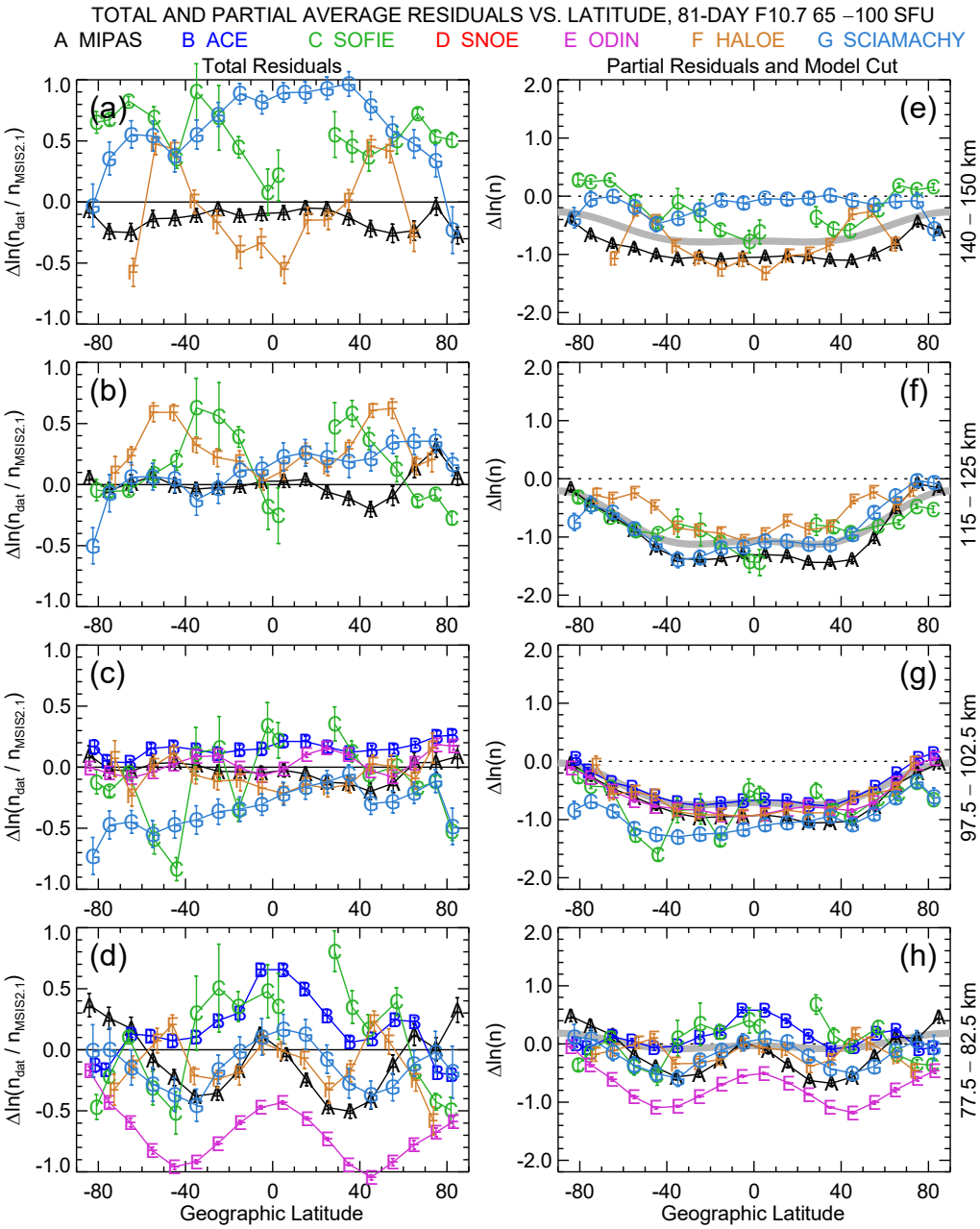

Figure S4. Same analysis as Figure 7 in the main text, but highlighting the latitude modulation of the $\bar{F}_{10.7}$ dependence. Results are shown for $\bar{F}_{10.7} \leq 100$, geomagnetically quiet ($Ap \leq 6$) conditions, and including data from all times of the year. In panels (e)–(h) on the right, the partial residuals were calculated relative to MSIS 2.1 with the $\bar{F}_{10.7}$ terms and the coupled $\bar{F}_{10.7}$-latitude terms turned off. The thick gray curve shows the latitude variation of the model due to the coupled $\bar{F}_{10.7}$-latitude terms, evaluated at the average $\bar{F}_{10.7}$ of each latitude bin and referenced to the latitudinally averaged (weighted by cosine latitude) model value at $\bar{F}_{10.7} \cong 150$.

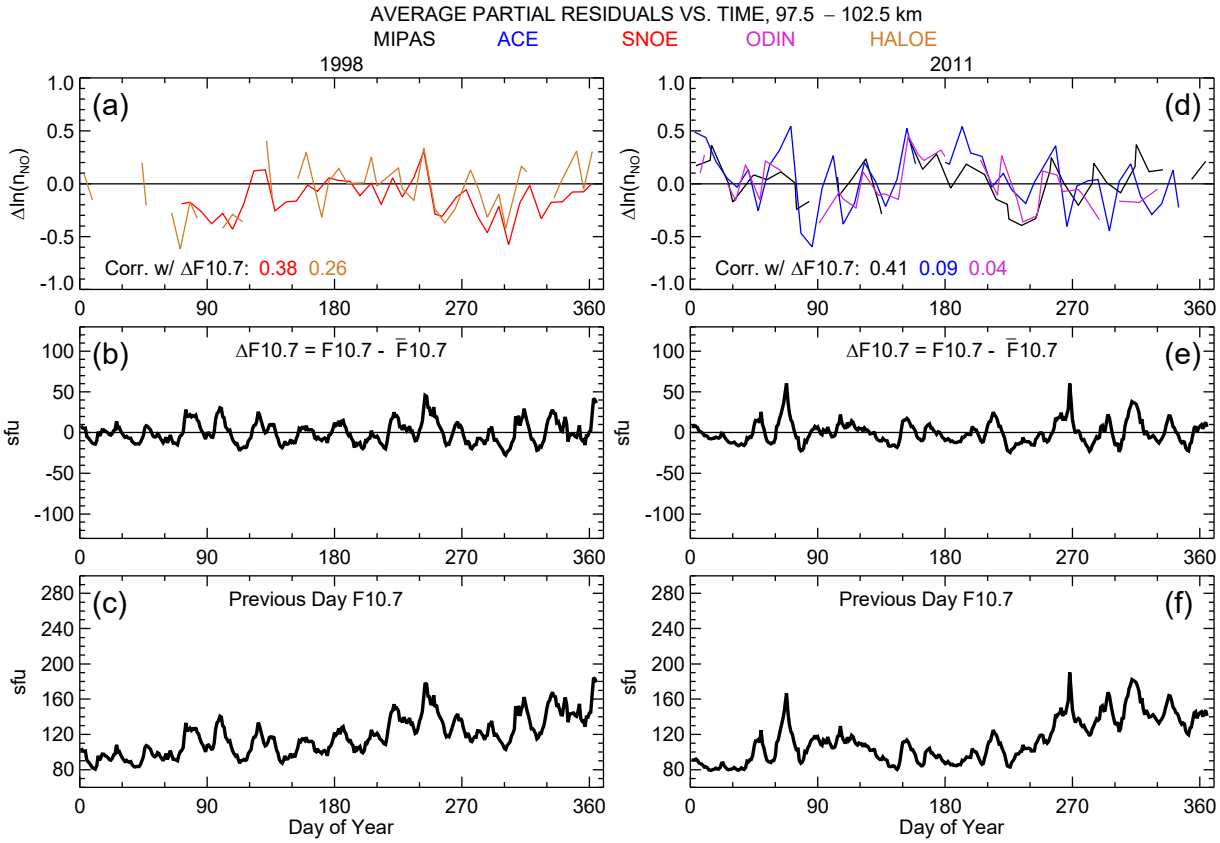

Figure S5. Average partial data-minus-model residuals, as a function of time, relative to MSIS 2.1 with the $\Delta F_{10.7}$ terms turned off, thereby highlighting the short-term variations in the data that are partly attributable to $F_{10.7}$ variations. Same as Figure 11 in the main text, but showing results for 1998 and 2012.

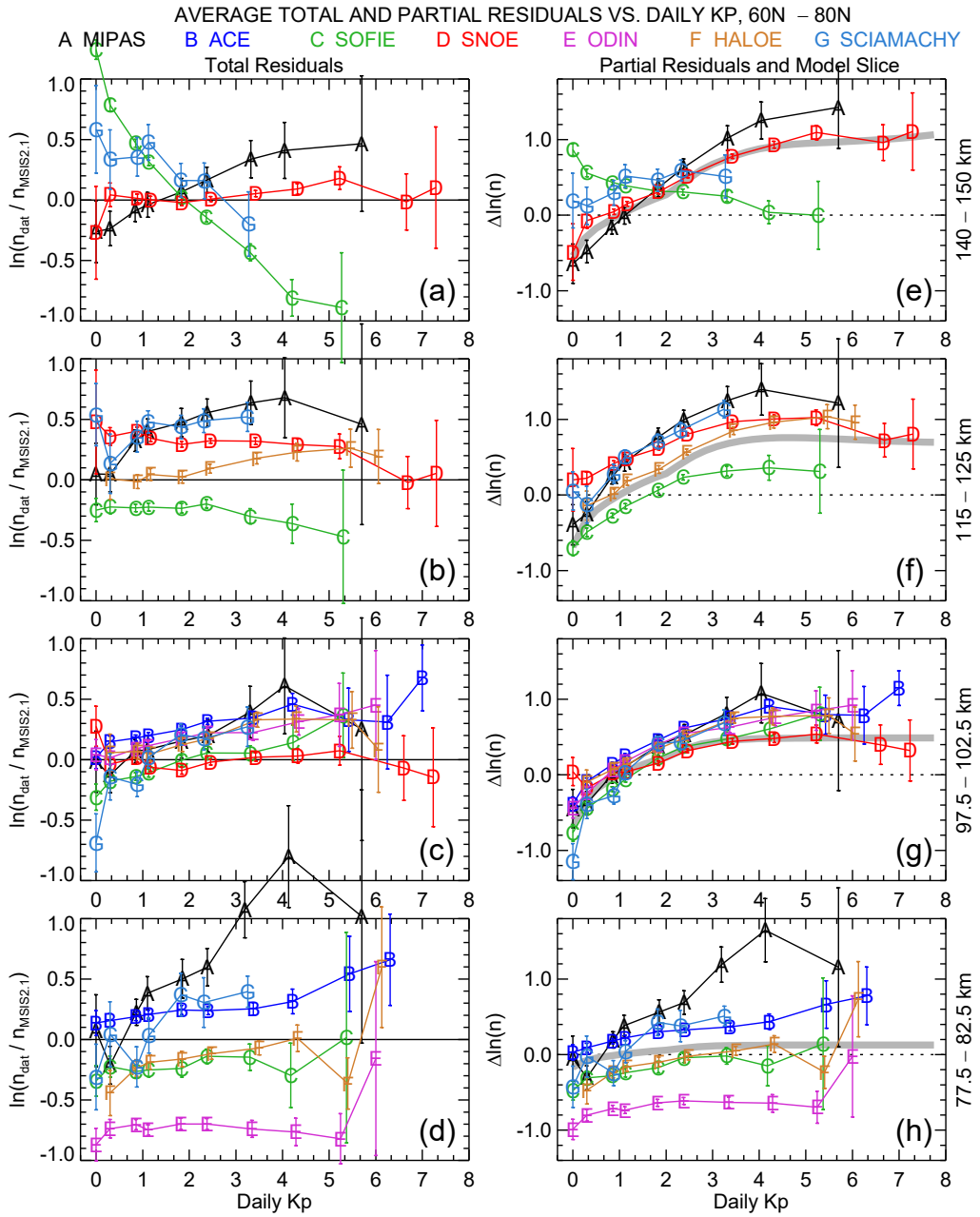

Figure S6. Total (left) and partial (right) data-minus-model residuals as a function of the daily average *Kp* index of geomagnetic activity. Same as Figure 12 in the main text, but showing results for high northern latitudes (60°N–80°N).

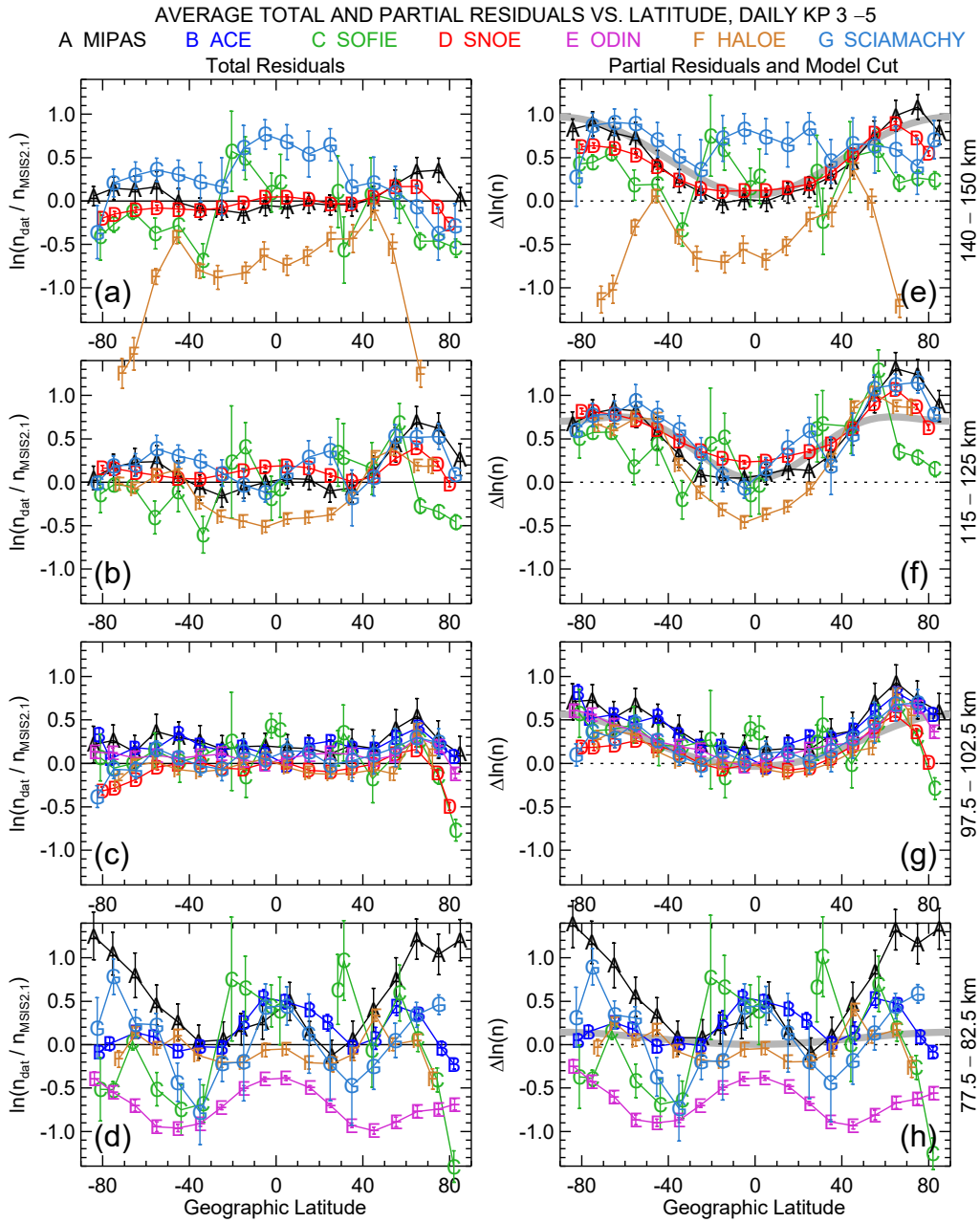

Figure S7. Same analysis as Figure 7 in the main text, but highlighting the latitude modulation of the geomagnetic activity dependence. Results are shown for $3 \leq$ daily $Kp \leq 5$, and including data from all times of the year and all $F_{10.7}$ levels. In panels (e)–(h) on the right, the partial residuals were calculated relative to MSIS 2.1 with the global Ap and the coupled Ap-latitude terms turned off. The thick gray curve shows the latitude variation of the model due to the coupled Ap-latitude terms, evaluated at the average Kp (and thence converted to Ap) of each latitude bin and referenced to the latitudinally averaged (weighted by cosine latitude) model value at $Ap \cong 4$.

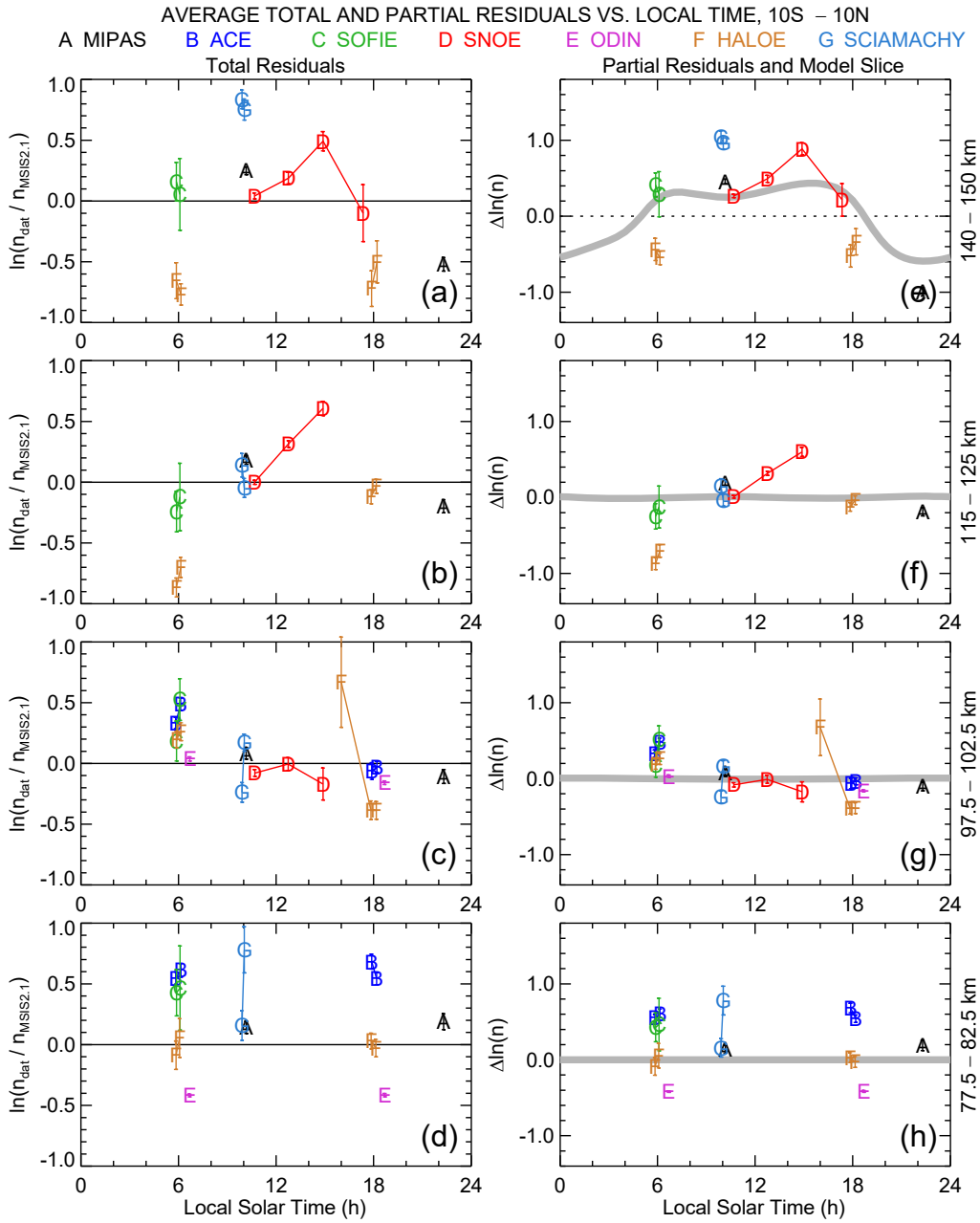

Figure S8. Same analysis as Figure 7 in the main text, but as a function of local time, near with equator (10°S–10°N) with all seasons combined. The local time bins are 2 hours wide (approximately monthly bins). In panels (e)–(h) on the right, the thick gray curve shows the local time variation of the model referenced to the diurnal average (dotted horizontal line).

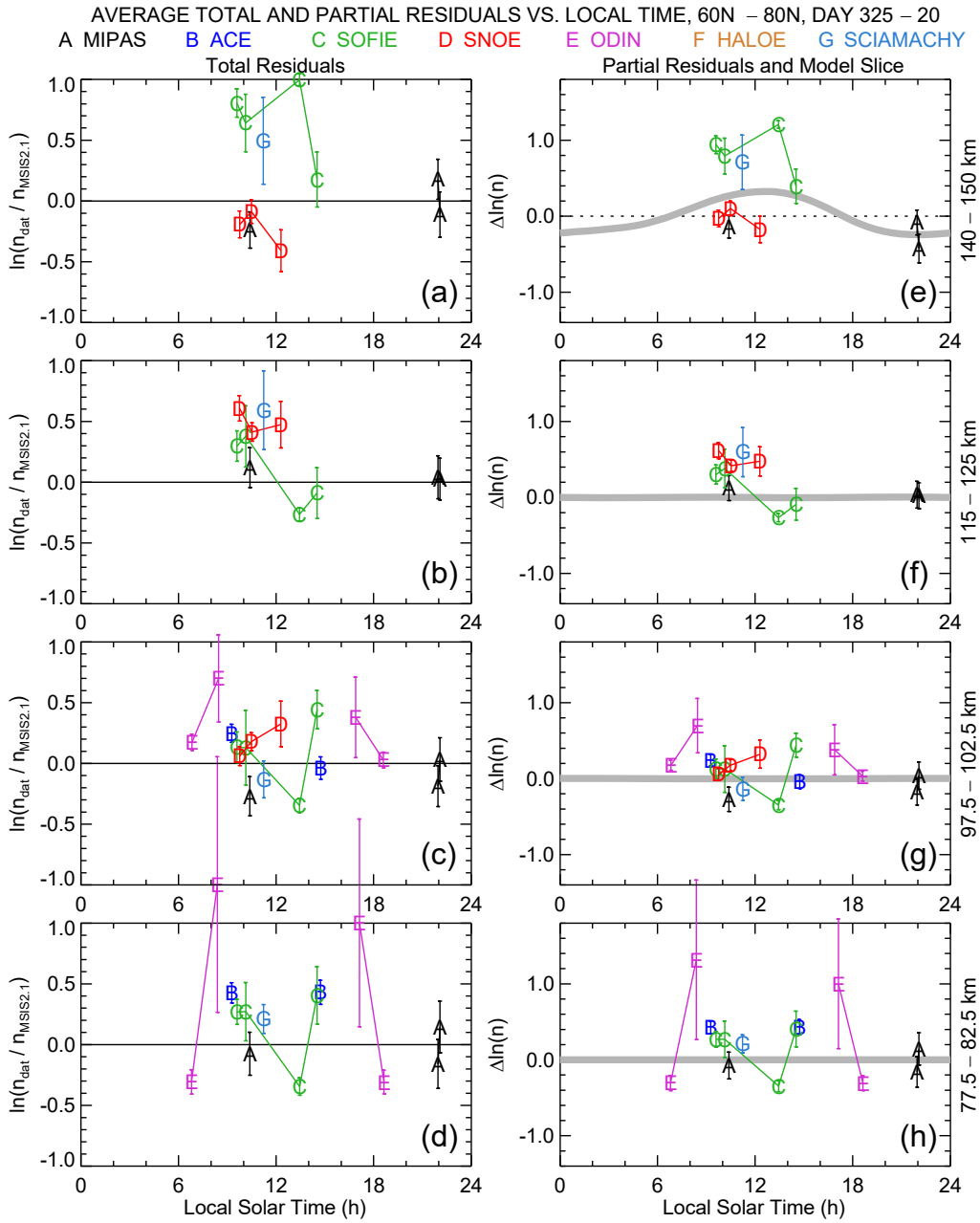

Figure S9. Same Figure S8, but showing results at northern high latitudes (60°N–80°N) during winter (day 325–20).

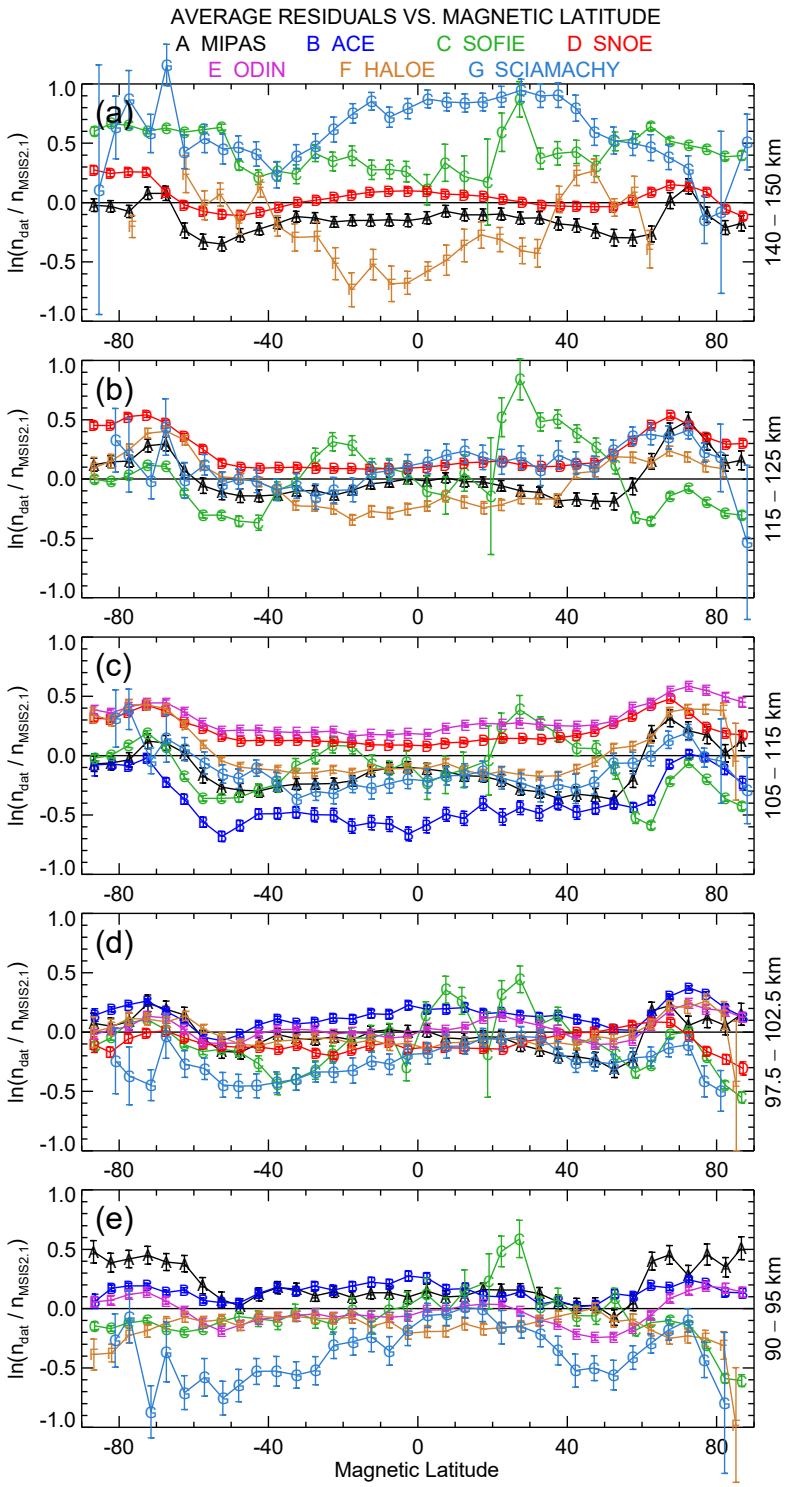

Figure S10. Average data-minus-model residuals as a function of Quasi-Dipole geomagnetic latitude (Richmond, 1995; Emmert et al., 2010), for geomagnetically quiet ($Ap \leq 6$) conditions and in the five altitude bins indicated to the right of each panel. All 15 analysis ensembles were included in the averages, along with all of the independent SCIAMACHY daily average data. Results from each data set are coded by color and letter, as indicated in the legend. The latitude bins are 5° wide, from 90°S to 90°N. Error bars indicate the 1σ estimated uncertainty of the mean.

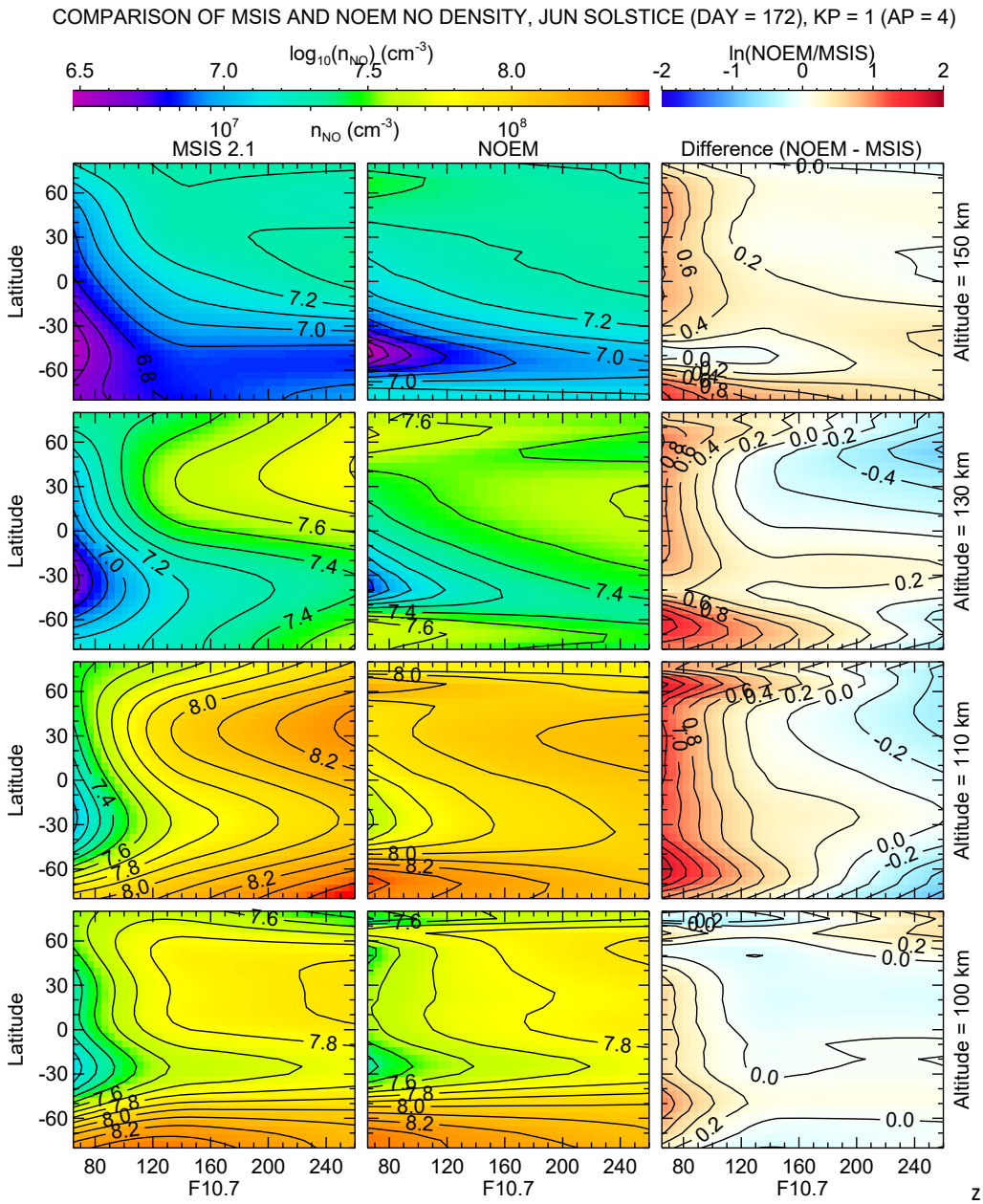

Figure S11. (left) MSIS 2.1 $\log_{10}$ NO density at 1100 local time and June solstice (day of year = 172), as a function of geographic latitude (80°S–80°N) and $F_{10.7}$ (65–260 sfu), at the altitudes indicated to the right of each row. The contour interval is 0.1 (~26% increase with each successively higher contour). (center) Corresponding NOEM (Marsh et al., 2004) density, as a function of *geomagnetic* latitude and $F_{10.7}$. (right) Natural log of the NOEM/MSIS ratio. The contour interval is 0.2 (~22%).

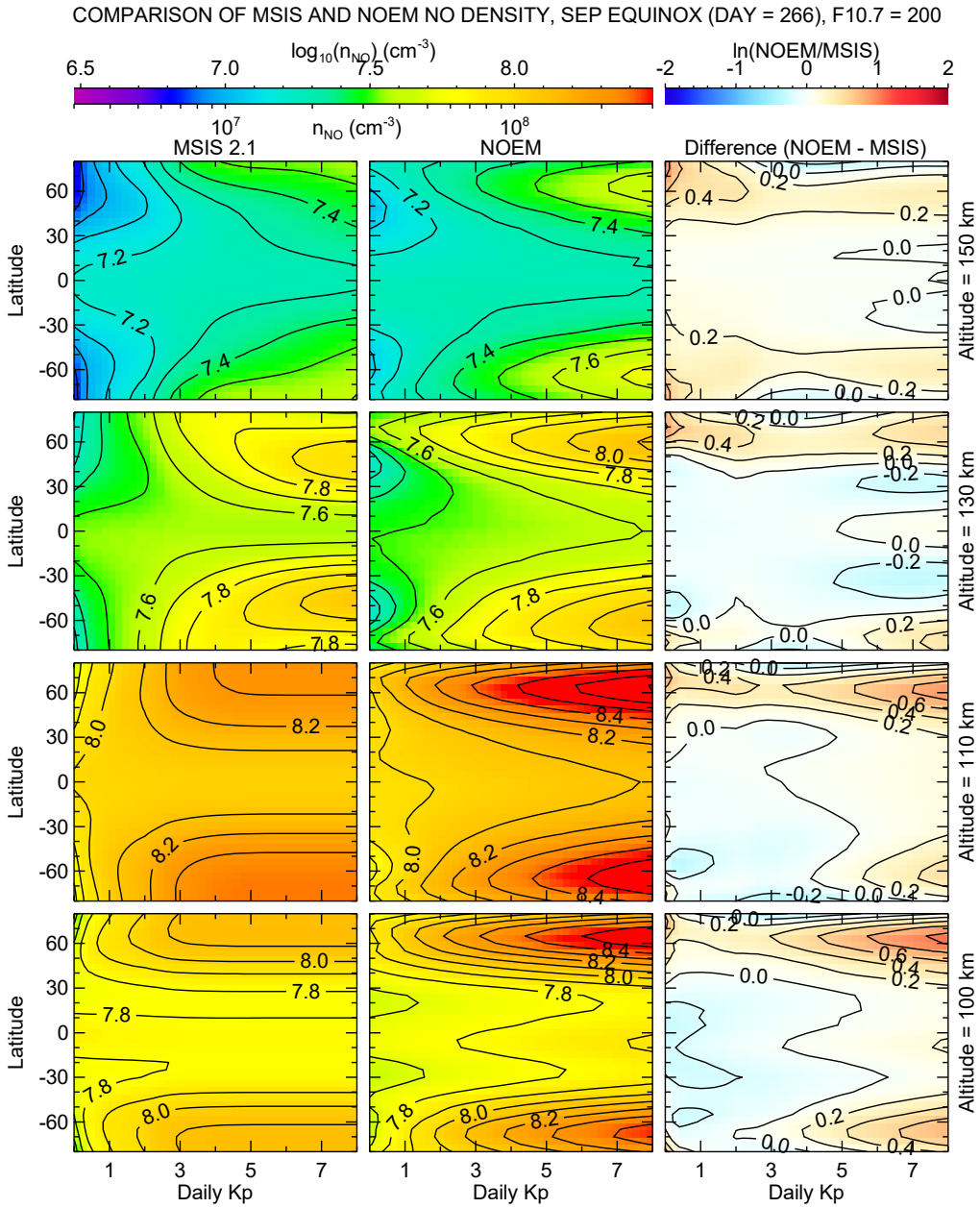

Figure S12. (left) MSIS 2.1 $\log_{10}$ NO density at 1100 local time, September equinox (day of year = 266), and $F_{10.7}$ = 200 sfu, as a function of geographic latitude (80°S–80°N) and daily *Kp* (0–8), at the altitudes indicated to the right of each row. The contour interval is 0.1 (~26% increase with each successively higher contour). (center) Corresponding NOEM (Marsh et al., 2004) density, as a function of *geomagnetic* latitude and *Kp*. (right) Natural log of the NOEM/MSIS ratio. The contour interval is 0.2 (~22%).

Data Set S1. (2022JA030896-sup-0004-Data Set SI-S01.xlsx) MSIS 2.1 NO Parameter Values. Each column of the spreadsheet corresponds to a different vertical profile parameter (described in the top row) from Equation 1, and each row corresponds to a different expansion parameter. When a parameter value is zero, the cell is blank. The first column ('iterm') contains the expansion parameter indices j. For the linear expansion terms (j = 0–383), the second column ('term') describes the terms to which the parameters are linearly applied. For the nonlinear expansion terms ($j \geq 384$), the second column ('term') lists the variables involved in the terms, the third column ('parameter') lists the parameter labels used in MSISE-00, and the fourth column ('term2') describes the terms that involve those parameters. This spreadsheet only includes the NO parameter values; the values for temperature and other species are unchanged from MSIS 2.0.

Data Set S2. (2022JA030896-sup-0005-Data Set SI-S02.xlsx) Same as Data Set S1, but containing initial parameter values tuned only to TIME-GCM and NOEM output. See main text for details.

Software S1. (2022JA030896-sup-0002-Supporting Information SI-S02.tar.gz or 2022JA030896-sup-0003-Supporting Information SI-S03.zip) NRLMSIS2.1 software distribution package (NRL-SOF-014-1). Use is governed by the Open Source Academic Research License Agreement contained in the file nrlmsis2.1_license.txt, which is included in the package.